

17-07-2026

Lab

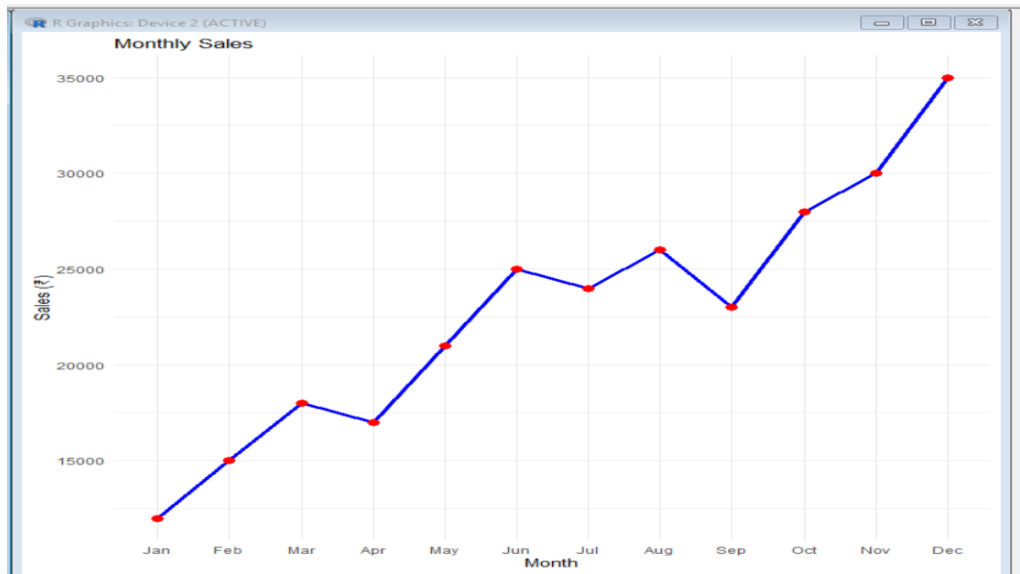
Set 1: Monthly Sales Data

Dataset: Monthly Sales

1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

CODE:

```
library(ggplot2)
sales$Month <- factor(
  sales$Month,
  levels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
    "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")
)
ggplot(sales, aes(x = Month, y = Sales, group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Monthly Sales",
    x = "Month",
    y = "Sales (₹)"
  ) +
  theme_minimal()
```



2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.

```
library(ggplot2)

# Total product sales
product_sales <- aggregate(Product_Sales ~ Top_Product,
                             data = sales,
                             sum)

# Bar Chart
ggplot(product_sales,
        aes(x = Top_Product,
            y = Product_Sales,
            fill = Top_Product)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Top Selling Products",
    x = "Product",
    y = "Total Units Sold"
  ) +
  theme_minimal()
```

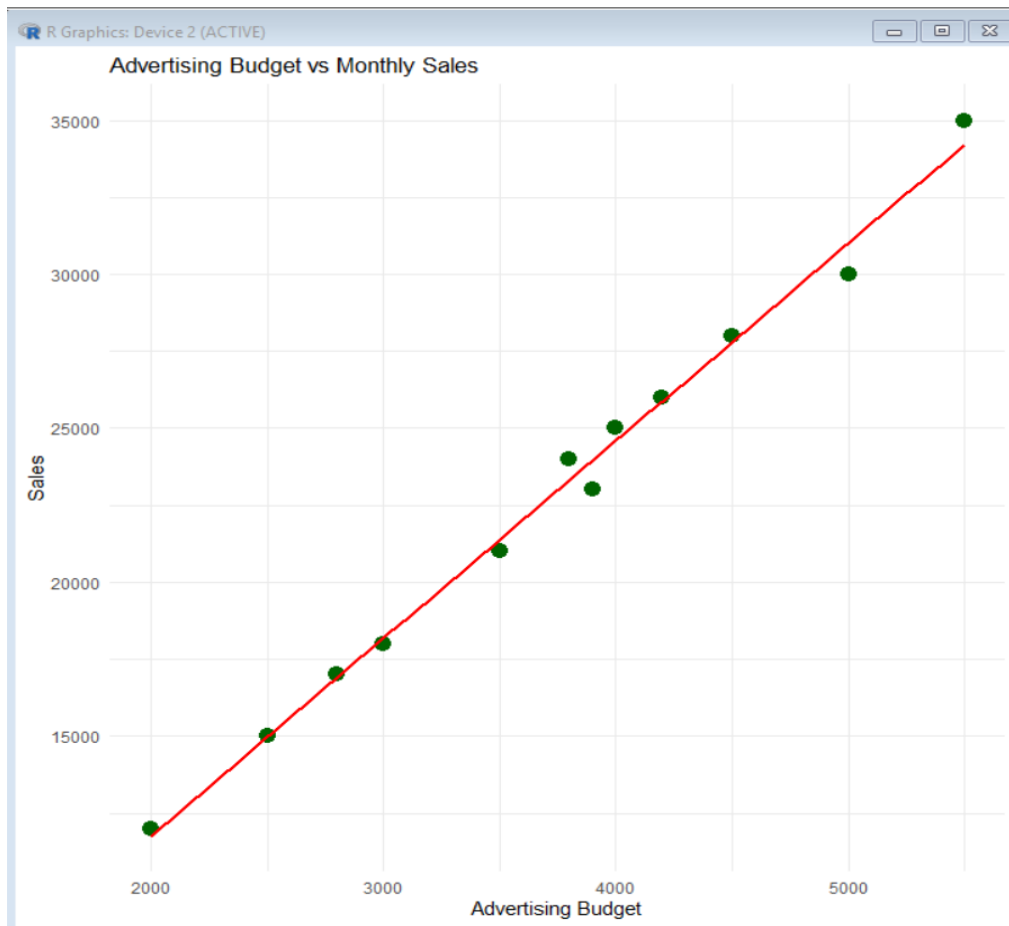


3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales.

Explain the insights drawn from the scatter plot.

```
library(ggplot2)
```

```
ggplot(sales,  
  aes(x = Advertising_Budget,  
    y = Sales)) +  
  geom_point(color = "darkgreen", size = 4) +  
  geom_smooth(method = "lm", se = FALSE, color = "red") +  
  labs(  
    title = "Advertising Budget vs Monthly Sales",  
    x = "Advertising Budget",  
    y = "Sales"  
  ) +  
  theme_minimal()
```

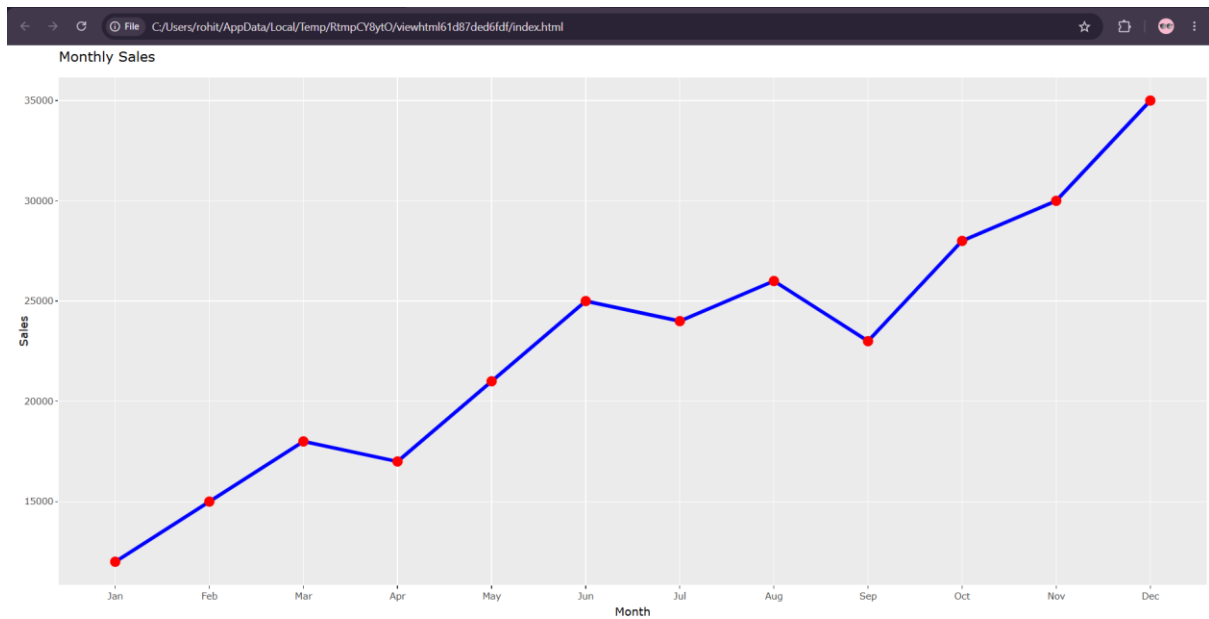


4. Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively.

Interactive Line Chart

```
line_plot <- ggplot(sales,
  aes(x = Month,
    y = Sales,
    group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Monthly Sales",
    x = "Month",
    y = "Sales"
  )

ggplotly(line_plot)
```



Interactive Bar Chart

```
bar_plot <- ggplot(product_sales,  
  aes(x = Top_Product,  
      y = Product_Sales,  
      fill = Top_Product)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Top Selling Products",  
    x = "Product",  
    y = "Units Sold"  
  )  
  
ggplotly(bar_plot)
```



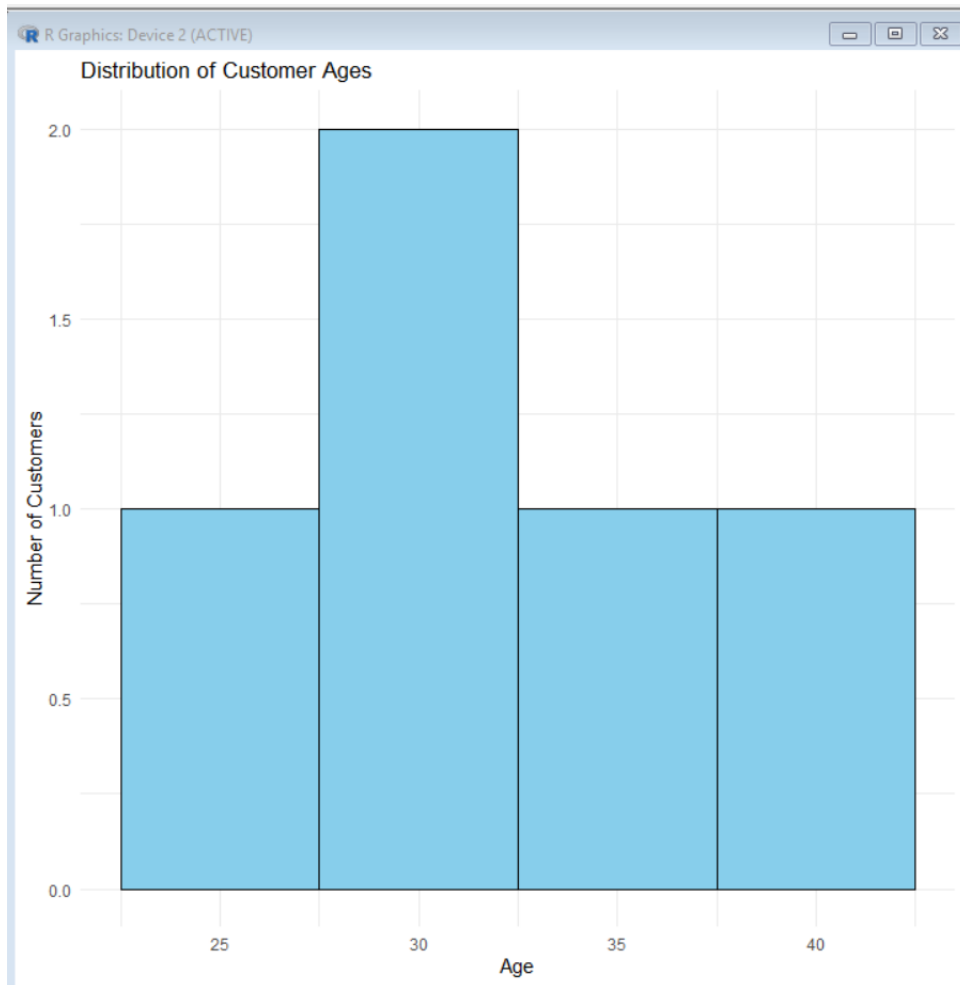
SET 2

1. Create a histogram to represent the distribution of customer ages. Label the axes and title the chart.

```
library(ggplot2)
```

```
customer <- read.csv(file.choose())
```

```
ggplot(customer, aes(x = Age)) +  
  geom_histogram(binwidth = 5,  
                 fill = "skyblue",  
                 color = "black") +  
  labs(  
    title = "Distribution of Customer Ages",  
    x = "Age",  
    y = "Number of Customers"  
  ) +  
  theme_minimal()
```



2. Generate a pie chart to display the overall distribution of customer satisfaction scores. Include labels.

```
customer <- read.csv(file.choose())
```

```
# Count each satisfaction score
```

```
score <- table(customer$Satisfaction_Score)
```

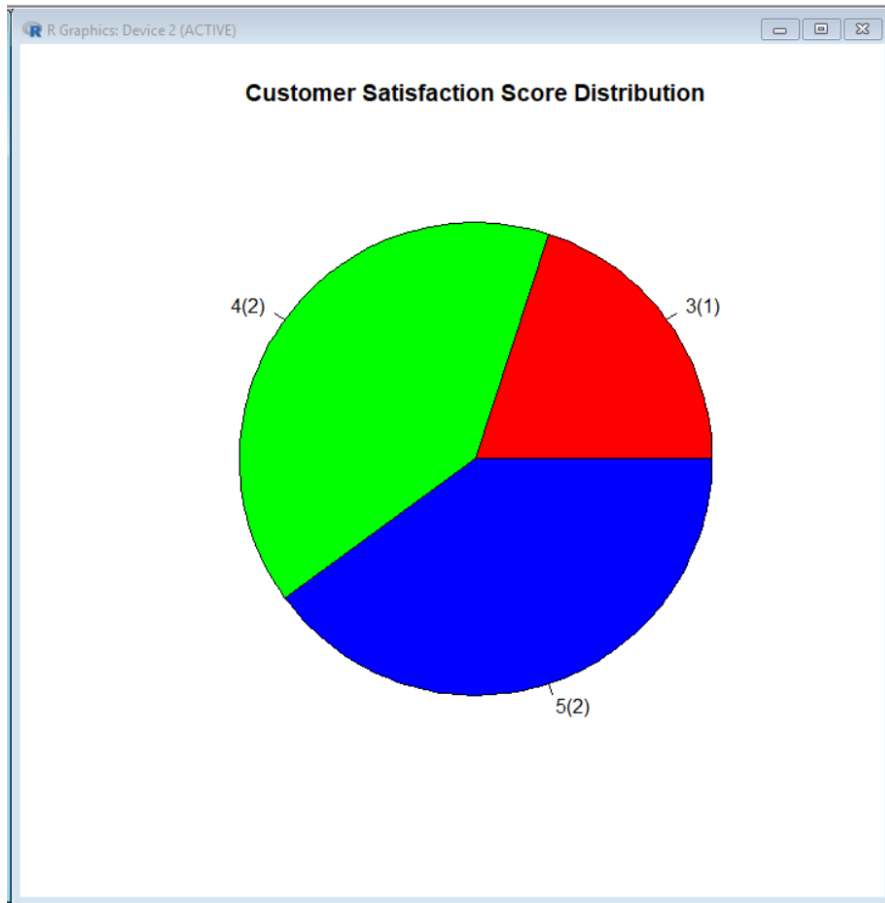
```
# Pie Chart
```

```
pie(score,
```

```
  labels = paste(names(score), "(", score, ")", sep=""),
```

```
  col = rainbow(length(score)),
```

```
  main = "Customer Satisfaction Score Distribution")
```



3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group.

```
library(ggplot2)
```

```
customer <- read.csv(file.choose())
```

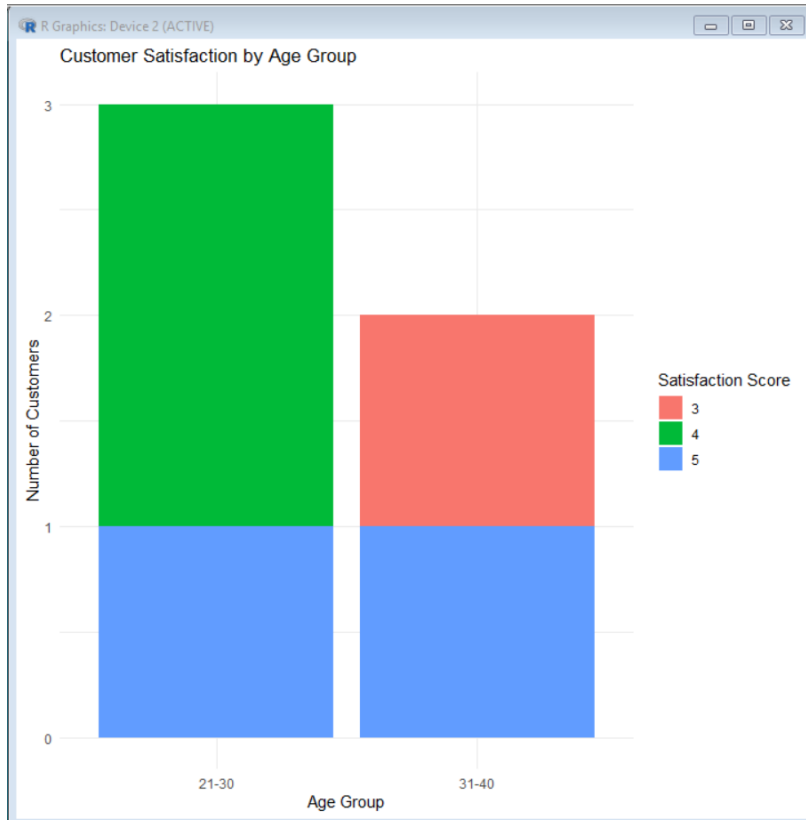
```
# Create Age Groups
```

```
customer$Age_Group <- cut(customer$Age,  
                           breaks = c(20,30,40,50),  
                           labels = c("21-30","31-40","41-50"),  
                           include.lowest = TRUE)
```

```
ggplot(customer,  
        aes(x = Age_Group,  
            fill = factor(Satisfaction_Score))) +  
geom_bar() +
```



```
labs(
  title = "Customer Satisfaction by Age Group",
  x = "Age Group",
  y = "Number of Customers",
  fill = "Satisfaction Score"
) +
theme_minimal()
```



4. Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments.

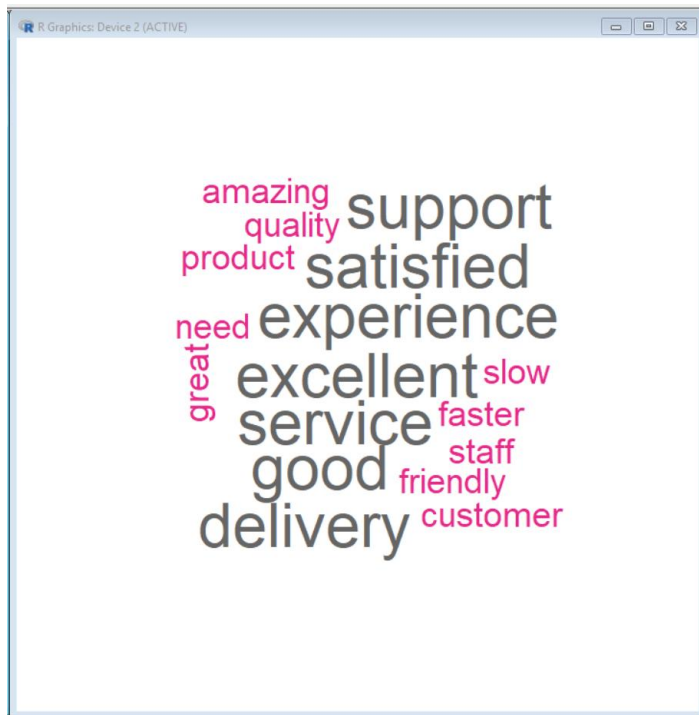
```
library(tm)
library(wordcloud)
library(RColorBrewer)

docs <- Corpus(VectorSource(feedback))

docs <- tm_map(docs, content_transformer(tolower))
docs <- tm_map(docs, removePunctuation)
docs <- tm_map(docs, removeNumbers)
```

```
docs <- tm_map(docs, removeWords, stopwords("english"))
```

```
wordcloud(docs,  
  max.words = 50,  
  random.order = FALSE,  
  colors = brewer.pal(8, "Dark2"))
```



Set 3: Employee Performance Evaluation

Dataset: Employee Performance

Employee ID Department Years of Service Performance Score

1 Sales 5 85

2 HR 3 92

3 Marketing 7 78

4 Sales 4 90

5 HR 2 76

1. Create a line chart to visualize the performance trend of employees over time. Include a legend and labels.

Dataset

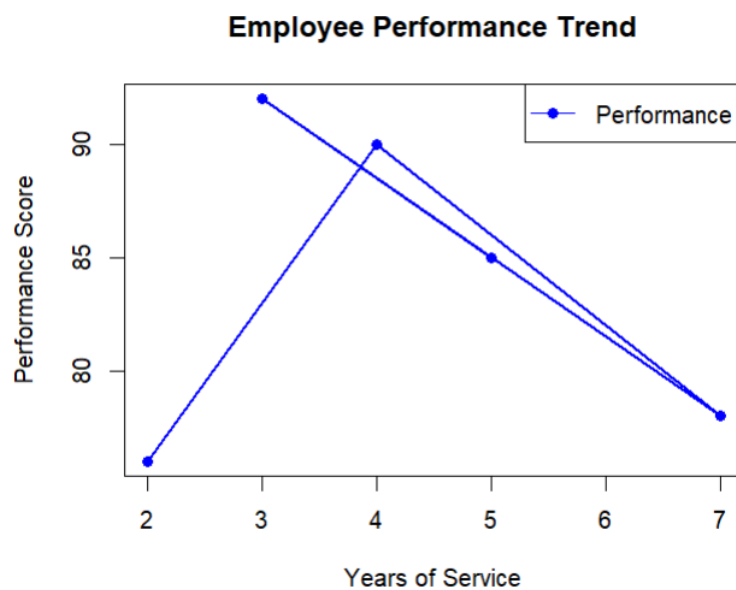
```
employee <- data.frame(
```

```

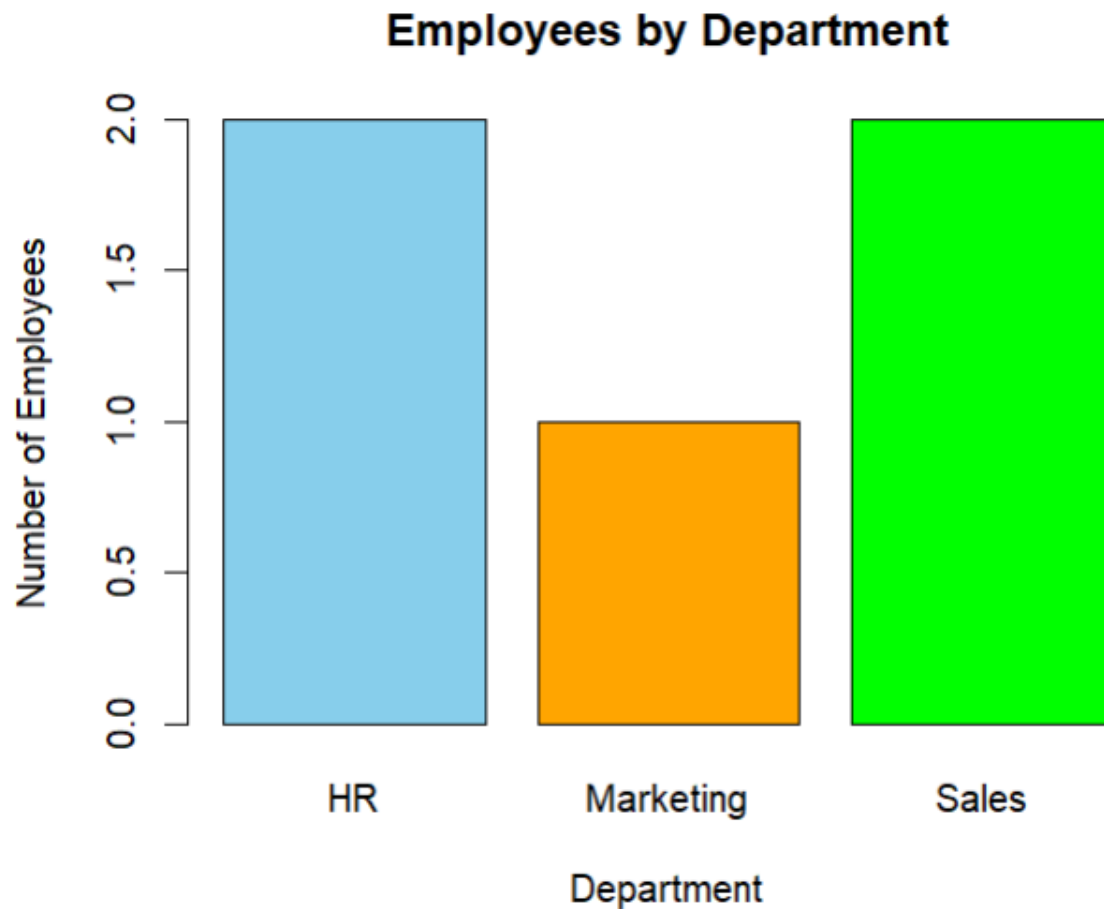
EmployeeID = c(1, 2, 3, 4, 5),
Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
YearsService = c(5, 3, 7, 4, 2),
Performance = c(85, 92, 78, 90, 76)
)

# Line Chart
plot(employee$YearsService,
      employee$Performance,
      type = "o",
      col = "blue",
      pch = 16,
      lwd = 2,
      xlab = "Years of Service",
      ylab = "Performance Score",
      main = "Employee Performance Trend")
legend("topright",
      legend = "Performance",
      col = "blue",
      lty = 1,
      pch = 16)

```



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.



3. Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

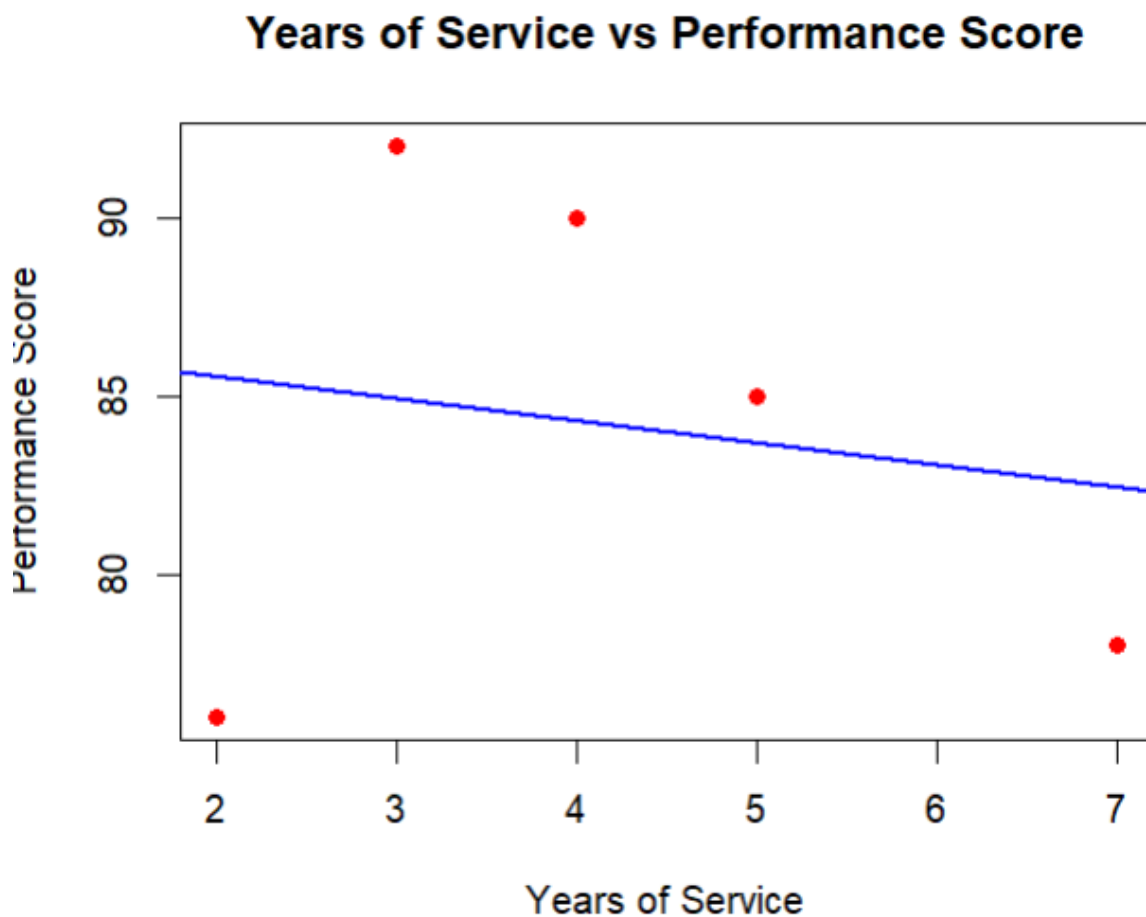
```
# Dataset
employee <- data.frame(
  EmployeeID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  YearsService = c(5, 3, 7, 4, 2),
  Performance = c(85, 92, 78, 90, 76)
)

# Scatter Plot
plot(employee$YearsService,
```

```
employee$Performance,  
pch = 19,  
col = "red",  
xlab = "Years of Service",  
ylab = "Performance Score",  
main = "Years of Service vs Performance Score")
```

```
# Regression Line
```

```
abline(lm(Performance ~ YearsService, data = employee),  
col = "blue",  
lwd = 2)
```



-
- 4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.**

```
library(shiny)
```

```
employee <- data.frame(  
  EmployeeID = c(1, 2, 3, 4, 5),  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  YearsService = c(5, 3, 7, 4, 2),  
  Performance = c(85, 92, 78, 90, 76)  
)
```

```
ui <- fluidPage(  
  

```

```
  titlePanel("Employee Performance Dashboard"),  
  

```

```
  sidebarLayout(  
  

```

```
    sidebarPanel(  
  

```

```
      selectInput("chart",  
        "Choose Chart:",  
        choices = c("Line Chart",  
          "Bar Chart"))  
  

```

```
    ),  
  

```

```
    mainPanel(  
  

```

```
      plotOutput("employeePlot")  
  

```

```
    )  
  

```

```
  )  
  

```

)

```
server <- function(input, output){
```

```
  output$employeePlot <- renderPlot({
```

```
    if(input$chart == "Line Chart"){
```

```
      plot(employee$YearsService,
            employee$Performance,
            type = "o",
            col = "blue",
            pch = 16,
            lwd = 2,
            xlab = "Years of Service",
            ylab = "Performance Score",
            main = "Employee Performance Trend")
```

```
      legend("topright",
            legend = "Performance",
            col = "blue",
            lty = 1,
            pch = 16)
```

```
    }else{
```

```
      dept_count <- table(employee$Department)
```

```
      barplot(dept_count,
            col = c("skyblue", "orange", "green"),
```

```

      xlab = "Department",
      ylab = "Number of Employees",
      main = "Employees by Department")

    }

  })

}

shinyApp(ui = ui, server = server)

```

Employee Performance Dashboard



Set 4: Product Inventory Management

Dataset: Product Inventory

Product ID Product Name Quantity Available

1 Product A 250

2 Product B 175

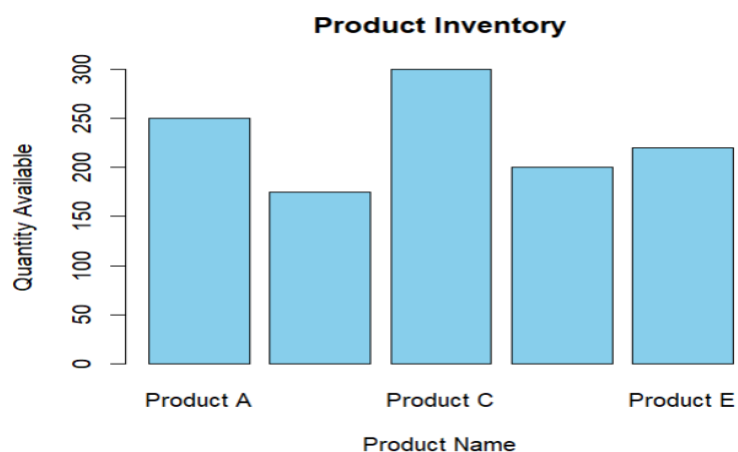
3 Product C 300

4 Product D 200

5 Product E 220

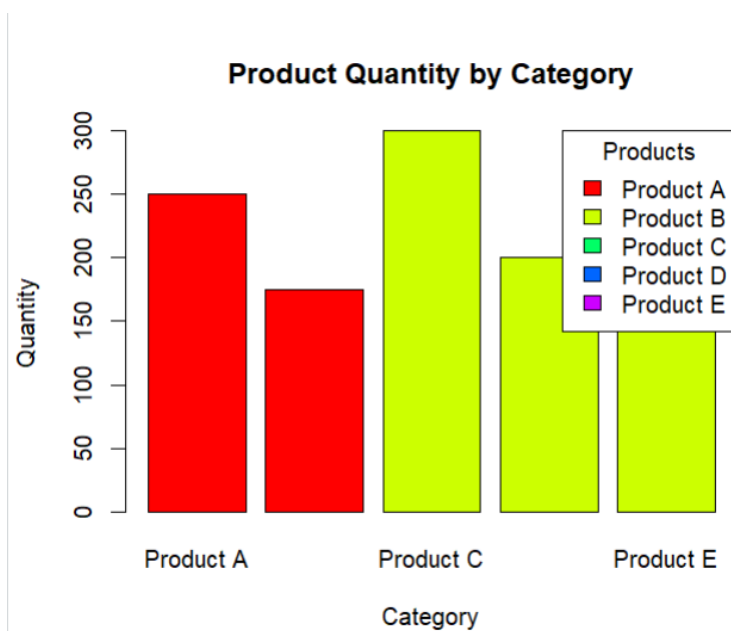
1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
inventory <- data.frame(  
  ProductID = c(1, 2, 3, 4, 5),  
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),  
  Quantity = c(250, 175, 300, 200, 220)  
)  
  
# Bar Chart  
  
barplot(inventory$Quantity,  
  names.arg = inventory$ProductName,  
  col = "skyblue",  
  xlab = "Product Name",  
  ylab = "Quantity Available",  
  main = "Product Inventory")
```



2. Generate a stacked bar chart to show the quantity of each product within different product categories.

```
inventory <- data.frame(  
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),  
  Quantity = c(250, 175, 300, 200, 220),  
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Furniture")  
)  
  
# Create Table  
table_data <- xtabs(Quantity ~ Category + ProductName, data = inventory)  
  
# Stacked Bar Chart  
barplot(table_data,  
  beside = FALSE,  
  col = rainbow(ncol(table_data)),  
  main = "Product Quantity by Category",  
  xlab = "Category",  
  ylab = "Quantity")  
  
legend("topright",  
  legend = colnames(table_data),  
  fill = rainbow(ncol(table_data)),  
  title = "Products")
```



3. Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

```
# Dataset
```

```
inventory <- data.frame(
```

```
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),
```

```
  Quantity = c(250, 175, 300, 200, 220),
```

```
  Price = c(100, 150, 80, 120, 90)
```

```
)
```

```
# Scatter Plot
```

```
plot(inventory$Price,
```

```
     inventory$Quantity,
```

```
     pch = 19,
```

```
     col = "red",
```

```
     xlab = "Product Price ($)",
```

```
     ylab = "Quantity Available",
```

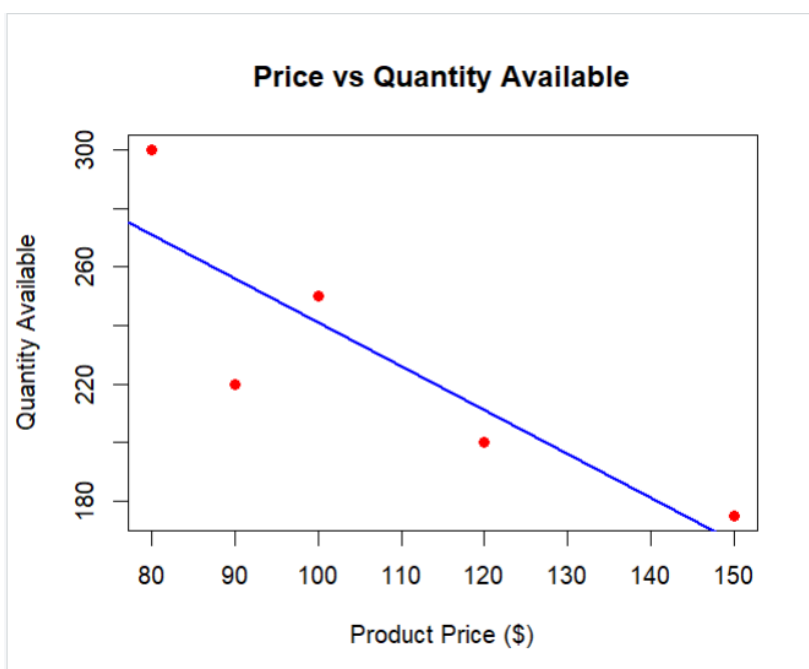
```
     main = "Price vs Quantity Available")
```

```
# Regression Line
```

```
abline(lm(Quantity ~ Price, data = inventory),
```

```
       col = "blue",
```

```
       lwd = 2)
```



4. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

```
library(shiny)

inventory <- data.frame(
  ProductID = c(1, 2, 3, 4, 5),
  ProductName = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Quantity = c(250, 175, 300, 200, 220),
  Category = c("Electronics", "Electronics", "Furniture", "Furniture", "Furniture")
)

ui <- fluidPage(
  titlePanel("Product Inventory Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput("chart",
        "Choose Chart:",
        choices = c("Bar Chart",
          "Stacked Bar Chart"))
    ),
    mainPanel(
      plotOutput("inventoryPlot")
    )
  )
)

server <- function(input, output){
  output$inventoryPlot <- renderPlot({
    if(input$chart == "Bar Chart"){
      barplot(inventory$Quantity,
        names.arg = inventory$ProductName,
        col = "skyblue",
        xlab = "Product Name",
        ylab = "Quantity Available",
```

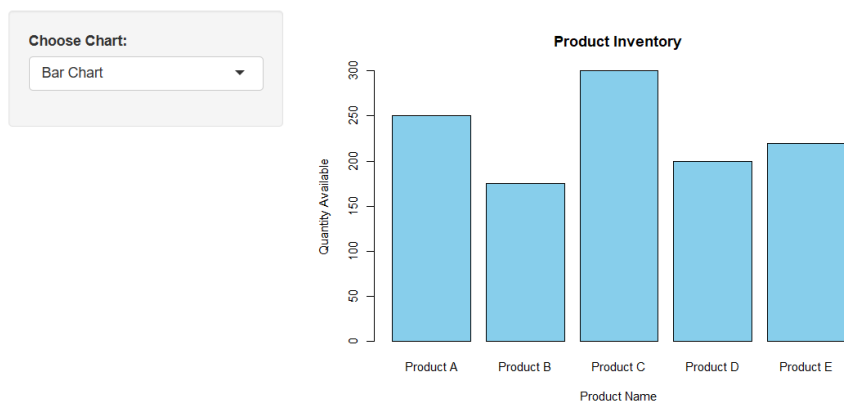
```

      main = "Product Inventory")
    }else{
      table_data <- xtabs(Quantity ~ Category + ProductName,
        data = inventory)
      barplot(table_data,
        beside = FALSE,
        col = rainbow(ncol(table_data)),
        main = "Product Quantity by Category",
        xlab = "Category",
        ylab = "Quantity")
      legend("topright",
        legend = colnames(table_data),
        fill = rainbow(ncol(table_data)),
        title = "Products")
    }
  })
}

shinyApp(ui = ui, server = server)

```

Product Inventory Dashboard



Set 5: Website Analytics

Dataset: Website Traffic

Date Page Views Click-through Rate

2023-01-01 1500 2.3%

2023-01-02 1600 2.7%

2023-01-03 1400 2.0%

2023-01-04 1650 2.4%

2023-01-05 1800 2.6%

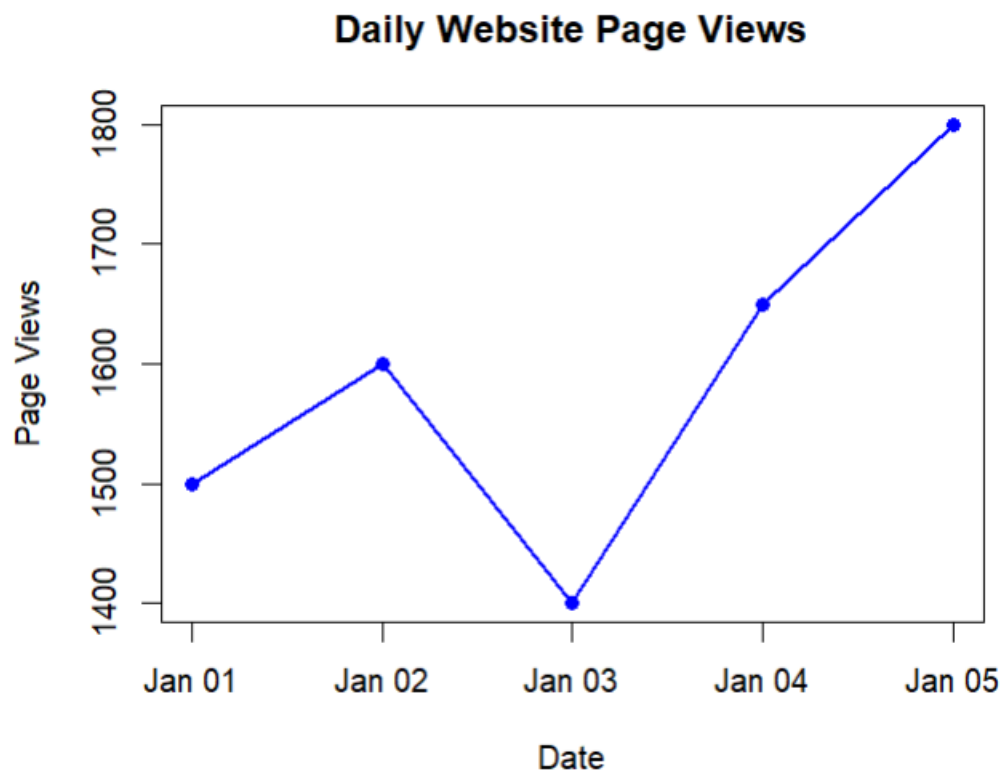
1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
# Dataset
```

```
traffic <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",  
                  "2023-01-04", "2023-01-05")),  
  PageViews = c(1500, 1600, 1400, 1650, 1800),  
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)  
)
```

```
# Line Chart
```

```
plot(traffic$Date,  
     traffic$PageViews,  
     type = "o",  
     col = "blue",  
     pch = 16,  
     lwd = 2,  
     xlab = "Date",  
     ylab = "Page Views",  
     main = "Daily Website Page Views")
```



2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
# Dataset
```

```
traffic <- data.frame(  
  Date = c("2023-01-01", "2023-01-02", "2023-01-03",  
            "2023-01-04", "2023-01-05"),  
  PageViews = c(1500, 1600, 1400, 1650, 1800),  
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)  
)
```

```
# Sort by CTR
```

```
traffic <- traffic[order(-traffic$CTR), ]
```

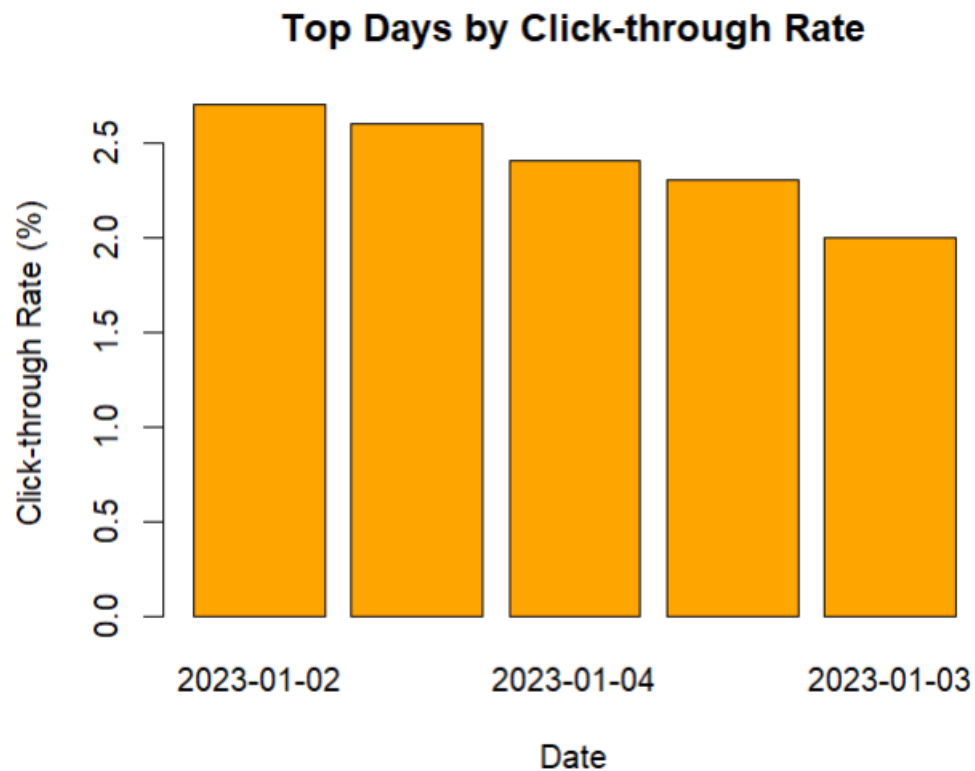
```
# Bar Chart
```

```
barplot(traffic$CTR,
```

```

names.arg = traffic$Date,
col = "orange",
xlab = "Date",
ylab = "Click-through Rate (%)",
main = "Top Days by Click-through Rate")

```



3. Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

Sample Dataset

```

traffic <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
    "2023-01-04", "2023-01-05")),
  Likes = c(200, 220, 180, 240, 260),
  Shares = c(80, 90, 70, 85, 95),
  Comments = c(40, 50, 35, 45, 55)
)

```



```

# Install package once

install.packages("ggplot2")

install.packages("reshape2")


library(ggplot2)

library(reshape2)

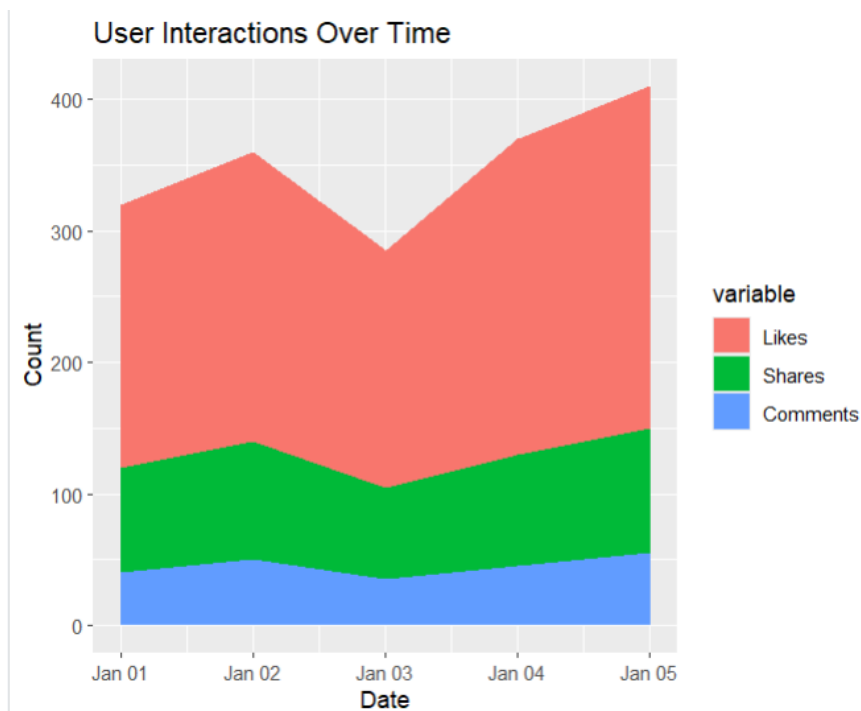

# Convert data to long format

traffic_long <- melt(traffic, id.vars = "Date")


# Stacked Area Chart

ggplot(traffic_long,
      aes(x = Date,
          y = value,
          fill = variable)) +
  geom_area() +
  labs(title = "User Interactions Over Time",
       x = "Date",
       y = "Count")

```



4. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

```
library(shiny)
```

```
traffic <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",  
    "2023-01-04", "2023-01-05")),  
  PageViews = c(1500, 1600, 1400, 1650, 1800),  
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)  
)
```

```
ui <- fluidPage(  
  
  titlePanel("Website Analytics Dashboard"),  
  
  sidebarLayout(  
  
    sidebarPanel(  
  
      selectInput("chart",  
        "Choose Chart:",  
        choices = c("Line Chart",  
          "Bar Chart"))  
  
    ),  
  
    mainPanel(  
  
      plotOutput("trafficPlot")  
  
    )  
  )  
)
```

```
)
```

```
)
```

```
server <- function(input, output){
```

```
  output$trafficPlot <- renderPlot({
```

```
    if(input$chart == "Line Chart"){
```

```
      plot(traffic$Date,  
            traffic$PageViews,  
            type = "o",  
            col = "blue",  
            pch = 16,  
            lwd = 2,  
            xlab = "Date",  
            ylab = "Page Views",  
            main = "Daily Website Page Views")
```

```
    }else{
```

```
      temp <- traffic[order(-traffic$CTR), ]
```

```
      barplot(temp$CTR,  
               names.arg = temp$Date,  
               col = "orange",  
               xlab = "Date",  
               ylab = "Click-through Rate (%)",  
               main = "Top Days by CTR")
```

```

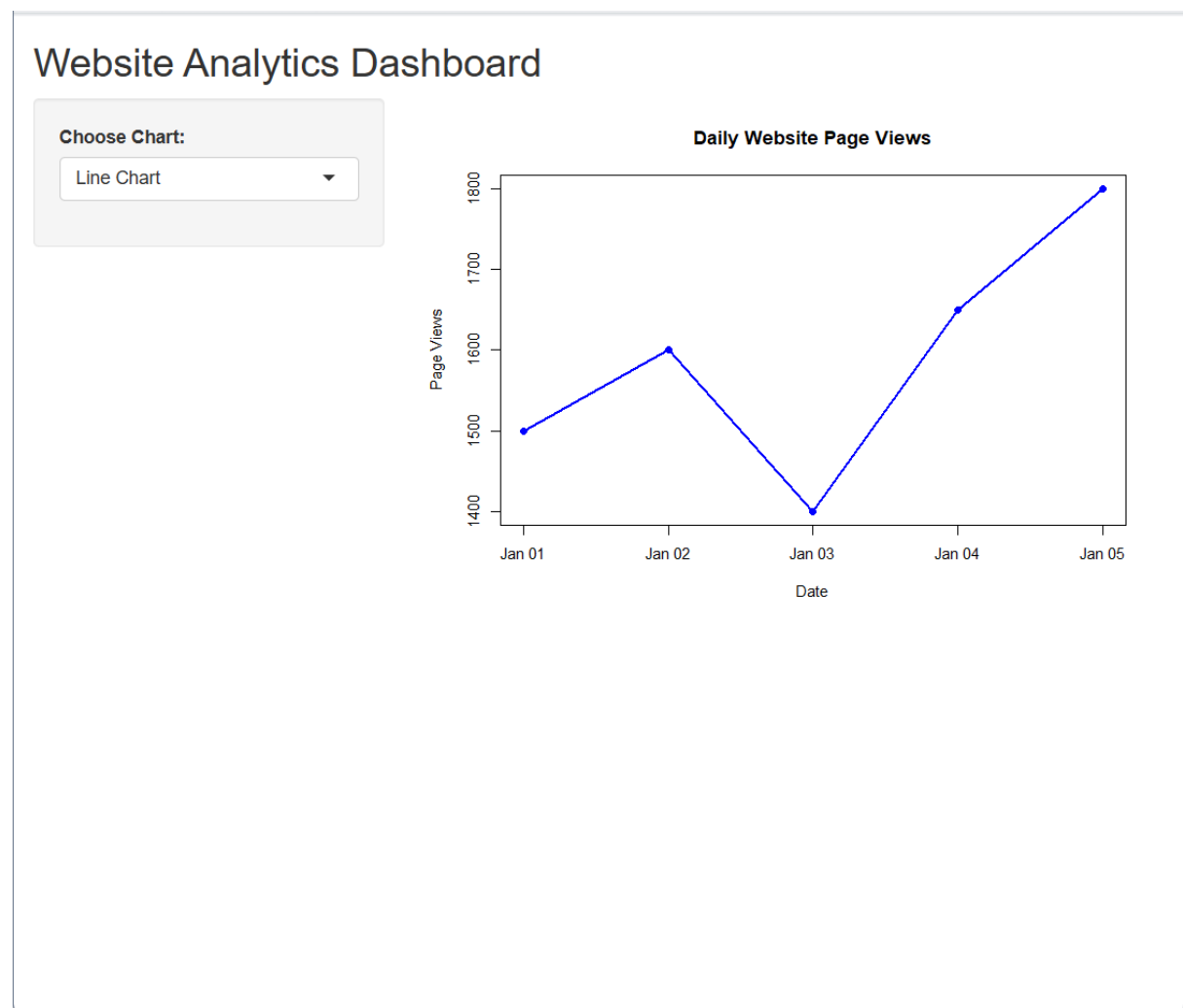
    }

  })

}

shinyApp(ui = ui, server = server)

```



Set 6

Product Sales Analysis

Dataset: Monthly Product Sales

Product ID Product Name January Sales February Sales

March

Sales

1 Product A 2000 2200 2400

2 Product B 1500 1800 1600

3 Product C 1200 1400 1100

- 1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.**

```
library(ggplot2)
```

```
library(reshape2)
```

```
# Convert data from wide format to long format
```

```
long <- melt(
```

```
  product_sales,
```

```
  id.vars = c("Product_ID", "Product_Name"),
```

```
  variable.name = "Month",
```

```
  value.name = "Sales"
```

```
)
```

```
# Create Grouped Bar Chart
```

```
ggplot(long,
```

```
  aes(x = Product_Name,
```

```
      y = Sales,
```

```
      fill = Month)) +
```

```
geom_bar(stat = "identity", position = "dodge") +
```

```
labs(
```

```
  title = "Quarter 1 Product Sales",
```

```
  x = "Product Name",
```

```
  y = "Sales"
```

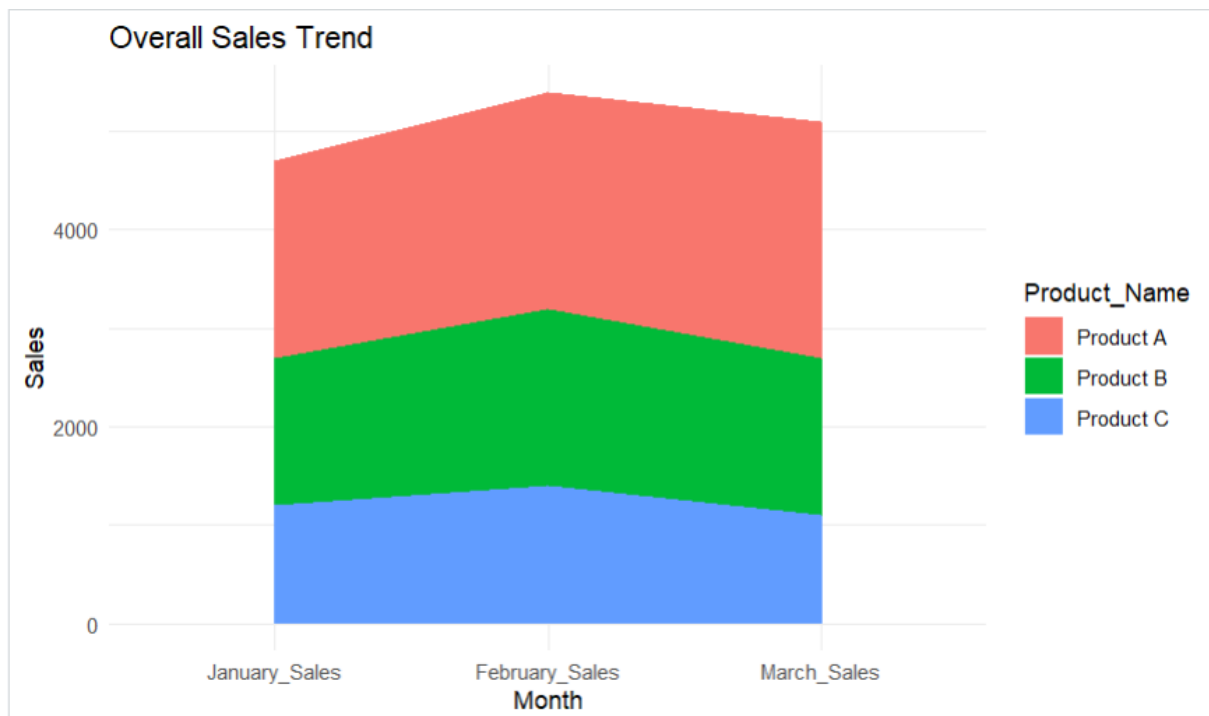
```
) +
```

```
theme_minimal()
```



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
ggplot(long,
  aes(x = Month,
    y = Sales,
    fill = Product_Name,
    group = Product_Name)) +
  geom_area() +
  labs(
    title = "Overall Sales Trend",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()
```



3. Build a table to show the monthly sales data for each product. Label the table elements.

```
print(product_sales)
```

```
> print(product_sales)
  Product_ID Product_Name January_Sales February_Sales March_Sales
1          1   Product A           2000           2200           2400
2          2   Product B           1500           1800           1600
3          3   Product C           1200           1400           1100
> |
```

4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

```
library(shiny)
```

```
traffic <- data.frame(
```

```
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                    "2023-01-04", "2023-01-05")),
```

```
  PageViews = c(1500, 1600, 1400, 1650, 1800),
```

```
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
```

```
)
```

```
ui <- fluidPage(
```

```
  titlePanel("Website Analytics Dashboard"),
```

```
  sidebarLayout(
```

```

sidebarPanel(
  selectInput("chart",
    "Choose Chart:",
    choices = c("Line Chart",
      "Bar Chart"))
),
mainPanel(
  plotOutput("trafficPlot")
)
)
)
server <- function(input, output){
  output$trafficPlot <- renderPlot({
    if(input$chart == "Line Chart"){
      plot(traffic$Date,
        traffic$PageViews,
        type = "o",
        col = "blue",
        pch = 16,
        lwd = 2,
        xlab = "Date",
        ylab = "Page Views",
        main = "Daily Website Page Views")
    }else{
      temp <- traffic[order(-traffic$CTR), ]
      barplot(temp$CTR,
        names.arg = temp$Date,
        col = "orange",
        xlab = "Date",
        ylab = "Click-through Rate (%)",
        main = "Top Days by CTR")
    }
  })
}

```



```
    }  
  })  
}  
shinyApp(ui = ui, server = server)
```

Website Analytics Dashboard

Choose Chart:

Line Chart ▼

